

Title

Data Flow Diagram – SB Stocks

Level 0 DFD

The SB Stocks system interacts primarily with two external entities:

1. User/Trader
2. Admin

User to System Data Flow

The user provides:

- Registration details.
- Login credentials.
- Stock search queries.
- Portfolio details.
- Buy requests.
- Sell requests.
- Strategy parameters.

The system returns:

- Authentication results.
- User dashboard information.
- Stock data.
- Portfolio information.
- Transaction history.
- Order information.
- Strategy analysis results.

Admin to System Data Flow

The admin provides:

- Authentication credentials.
- Administrative requests.

The system returns:

- User information.
- Transaction information.
- Order information.

- Application statistics.
- Stock data.

Level 1 DFD

Process 1: User Authentication

Input:

- Registration information.
- Login credentials.

Process:

- Validate user data.
- Hash passwords.
- Verify credentials.
- Generate authentication token.

Database:

- Users Collection.

Output:

- Registration response.
- Authentication token.
- User information.

Process 2: Stock Management

Input:

- Stock search query.
- Stock symbol.

Process:

- Search stocks.
- Retrieve stock details.
- Retrieve historical data.

Database:

- Stocks Collection.

Output:

- Stock listings.
- Stock details.
- Historical price information.

Process 3: Portfolio Management

Input:

- Portfolio name.
- Portfolio update request.

Process:

- Create portfolio.
- Retrieve portfolios.
- Update portfolio.
- Calculate portfolio performance.

Database:

- Portfolios Collection.

Output:

- Portfolio details.
- Holdings.
- Profit/loss information.

Process 4: Trading

Input:

- Stock symbol.
- Quantity.
- Portfolio ID.
- Buy or sell action.

Process:

- Validate request.
- Fetch stock price.
- Check balance or holdings.
- Update portfolio.
- Update virtual balance.

Database:

- Users.
- Stocks.
- Portfolios.
- Transactions.
- Orders.

Output:

- Transaction result.
- Updated portfolio.
- Updated balance.

Process 5: Admin Monitoring

Input:

- Admin request.

Process:

- Validate admin authorization.
- Retrieve system data.

Database:

- Users.
- Stocks.
- Orders.
- Transactions.

Output:

- Dashboard statistics.
- Users list.
- Orders list.
- Transactions list.